Payman Tohidifar

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Summary

Computational biologist and biomolecular engineer with 14+ years of experience spanning molecular biology, computational science, and AI to solve complex biological problems. Hands-on experience spans software development and AI — from conventional ML to modern architectures and foundational models — on the computational side, and molecular biology, genetics, and biosynthesis on the experimental side. A collaborative team player and independent, innovative problem-solver (6 patents filed), comfortable navigating shifting project scope across computational, wet-lab, and engineering domains. Passionate about applying AI to advance biological discovery.

Technical Skills

Programming/Scripting: Python, R, Bash, C++

Deep Learning Frameworks: PyTorch, JAX/Flax, TensorFlow/Keras (familiar)

Data Science & Computing: NumPy, SciPy, Pandas, Scikit-learn, Seaborn, Jupyter Lab, HPC/Slurm

SWE Tools: VS Code, Claude Code, Pixi/Conda, uv/pip, Docker, Git, CI/CD (GitLab CI, GitHub Actions)

CompBio & Bioinformatics: DESeq2, HMMER, BLAST, fastp, SeqKit, Bowtie2, Samtools, bedtools, IGV, EPI2ME, MEME Suite, Nextflow; Databases: NCBI, Ensembl, UniProt, AlphaFold, PDB, KEGG, GO

MolBio & Genetics: Protein engineering/expression, CRISPR editing, ONT sequencing, Flow cytometry, Genetic assembly

Experience

Scientist — Feb 2022 – Present
BP (Biosciences, Technology and Innovation) — *San Diego, CA*

- Engineered and deployed an end-to-end, ML-assisted low-N protein engineering pipeline — combining unsupervised (VAE, transformer) feature extraction with supervised fitness models — to improve industrial enzyme activity and thermostability.
- Built statistical and ML pipelines to analyze complex, multi-dimensional biorefinery data, directly informing key business decisions.
- Built a full-stack GUI for ingestion, consolidation, statistical analytics, and visualization of multi-modal biological data across projects; selected as a top-3 platform for executive leadership review (see *a demo prototype*).
- Co-developed a semi-automated HTP directed-evolution and protein-expression pipeline that cut turnaround time from 6 weeks to 2 weeks, accelerating the DBTL cycle.
- Developed an HTP platform (a novel FACS + 2A-peptide co-translation method) to boost protein production in an industrial fungal host, identifying multiple enzyme hyperproducers for scaled testing and cutting discovery time from 1+ year to 3 months.
- Standardized in-house Oxford Nanopore long-read sequencing and de novo genome assembly SOPs (EPI2ME, Nextflow).
- Led a 7-person cross-functional project on biosynthesis of value-added small molecules, resulting in publication of 6+ patent applications within 3 years.
- Built a CRISPR-MAD7 platform for single-nucleotide-resolution genetic engineering, enabling construction of 50+ bacterial cell lines and replacing a technology costing over $10K/yr.

Postdoctoral Research Associate — Aug 2019 – Feb 2022
Institute for Genomic Biology, UIUC — *Urbana, IL*

- Co-developed a modular genetic toolkit for multi-gene expression and CRISPR-Cas9 chromosomal editing in two non-model oleaginous yeasts, enabling biosynthesis of value-added small molecules (2 publications, 1 conference paper).

Graduate Research Assistant — Jan 2012 – May 2019
Department of Chemical & Biomolecular Engineering — *Urbana, IL*

- Co-developed a modular C++ framework for multiscale simulation of cellular population behavior in response to complex spatiotemporal chemical gradients (see *here* for details).
- Led three independent projects elucidating molecular mechanisms of unconventional chemoreceptor sensing in bacteria — bidirectional pH, alcohol, and toxic aromatic molecule sensing — spanning protein engineering, genetic engineering, bioinformatics, structural biology, and MD simulations; resulted in 3 journal publications and multiple conference presentations.
- Led discovery and characterization of an unconventional DNA-sensing receptor in *B. subtilis* using in vivo/in vitro assays and in silico motif-discovery analysis; presented at multiple conferences and awarded Best Presentation (see *here* for details).

Voluntary Projects

Deep Learning for Protein Function Prediction | *GitHub Repo* June 2026 – July 2026

- Built a modular deep learning platform (PyTorch, JAX/Flax) on the CAFA benchmark to predict multi-label protein function across 291 Gene Ontology terms from sequence alone, integrating ESM-2 transformer embeddings into an extensible framework for rapid experimentation with custom architectures, loss functions, and metrics.

Production ready GPT-2 Text Classifier From Scratch | *GitHub Repo* April 2026 – May 2026

- Built a GPT-2 architecture from scratch in PyTorch for text classification, with a configuration-driven pipeline that ingests multi-source, two-column datasets with zero code changes.
- Scaled training across multiple GPUs via Distributed Data Parallel (DDP) with real-time tracking in Weights & Biases (WandB), and built an end-to-end evaluation suite for benchmarking and iterative experimentation.

Lung Cancer Bulk RNA-Seq Data Analysis Pipeline | *GitHub Repo* Feb 2026 – April 2026

- Built an end-to-end Python/R pipeline for CCLE bulk RNA-seq data — count processing, PCA, hierarchical clustering, differential expression, and pathway enrichment.

Publications

Co-first authors are denoted with asterisks.

8. Koh, HG.*, **Tohidifar, P.***, Oh, H., Ye, Q., Jung, SC., Rao, CV., Jin, YS. (2025) RT-EZ: A Golden Gate Assembly Toolkit for Streamlined Genetic Engineering of *Rhodotorula toruloides*. ACS Synth. Biol.

7. Hamed, S., Emara, M., **Tohidifar, P.**, Rao, CV. (2025) N-Acetyl Cysteine Exhibits Antimicrobial and Anti-Virulence Activity Against *Salmonella enterica*. PLoS One.

6. Yook, S., Deewan, A., Ziolkowski, L., Lane, S., **Tohidifar, P.**, Cheng, M., Singh, V., Stasiewicz, MJ., Rao, CV., Jin, YS. (2025) Engineering and Evolution of *Yarrowia lipolytica* for Producing Lipids from Lignocellulosic Hydrolysates. Bioresource Technology.

5. Bodhankar, GA., **Tohidifar, P.**, Foust, ZL., Ordal, GW., Rao, CV. (2022) Characterization of Opposing Responses to Phenol by *Bacillus subtilis* Chemoreceptors. J Bacteriol.

4. Joseph, E.*, **Tohidifar, P.***, Sarver, CT., Mackie, RI., Rao, CV. (2022) Fundamentals of Polymer Biodegradation Mechanisms. Wiley (Book chapter).

3. **Tohidifar, P.**, Bodhankar, GA.*, Pei, S.*, Cassidy, CK., Walukiewicz, HE., Ordal, GW., Stansfeld, PJ., Rao, CV. (2020) The Unconventional Cytoplasmic Sensing Mechanism for Ethanol Chemotaxis in *Bacillus subtilis*. mBio.

2. **Tohidifar, P.**, Plutz, MJ., Ordal, GW., Rao, CV. (2020) The Mechanism of Bidirectional pH Taxis in *Bacillus subtilis*. J Bacteriol.

1. Walukiewicz, HE., **Tohidifar, P.**, Ordal, GW., Rao, CV. (2014) Three Adaptation Systems of *B. subtilis* Chemotaxis. Molecular Microbiology.

Patents

4. "Recombinant microorganisms for beta-myrecene and nerol production." Valle, F., Vu, HNN., **Tohidifar, P.** Patent Application Pub. No. WO/2026/122843 A1, published June 11, 2026.

3. "Engineered dxp pathway for improved isoprenoid production in *E. coli*." Valle, F., **Tohidifar, P.**, Najdi, T., Vu, HNN. Patent Application Pub. No. WO/2025/155822 A1, published July 24, 2025.

2. "Recombinant microorganisms with increased accumulation and/or flux of cytidine triphosphate (CTP)." Valle, F., **Tohidifar, P.**, Najdi, T. Patent Application Pub. No. WO/2025/111429 A1, published May 30, 2025.

1. "Gluconate-regulated expression systems." Valle, F., **Tohidifar, P.** Patent Application Pub. No. WO/2025/059030 A1, published March 20, 2025.

- Co-invented 2 provisional patent applications with colleagues, covering novel enzymes and metabolic pathways for the biosynthesis of value-added small molecules (filed Feb 2 and Feb 19, 2026).

Education

University of Illinois at Urbana-Champaign (UIUC) Urbana, IL
Ph.D. in Chemical & Biomolecular Engineering *2019*

- Computational Science & Engineering Certificate, 2019
- Dissertation: *Non-canonical sensing mechanisms in bacteria*

Sharif University of Technology Tehran, Iran
B.S. in Chemical Engineering *2011*

Coursework & Certifications

- Self-Study: *Deep Learning for Biology*, C. Ravarani and N. Latysheva (O'Reilly Media, 2025) — in progress (*Notes*)
- *Claude Code: A Highly Agentic Coding Assistant*, DeepLearning.AI (2026)
- Deep Learning Specialization, Coursera (2026) (*Credential*, *Notes*)
- Self-Study: *Build a Large Language Model (From Scratch)*, S. Raschka (Manning Publications, 2024) — completed 2026 (*Notes*)
- Machine Learning Specialization, Coursera (2025) (*Credential*, *Notes*)
- Mathematics for Machine Learning and Data Science Specialization, Coursera (2025) (*Credential*)
- bp Data Science Intermediate Bootcamp, Pragmatic Institute (2024) (*Credential*)
- ChIP-Seq Analysis Hands-On Training, High-Performance Biological Computing, UIUC (2017)

Selected Conference Talks

- **Tohidifar, P.**, Kimura, Y., Rao, CV. (2015) A modular computational framework for multi-scale simulation of chemotaxis in complex environments. AIChE, Salt Lake City, UT.
- **Tohidifar, P.**, Walukiewicz, HE., Ordal, GW., Rao, CV. (2016) DNA taxis in *Bacillus subtilis*. Receptor Fest 19th Annual Meeting, Santa Barbara, CA.
- **Tohidifar, P.**, Walukiewicz, HE., Ordal, GW., Rao, CV. (2017) DNA sensing in *Bacillus subtilis*. BLAST XIV, New Orleans, LA.

Honors & Awards

- Nucleic Acids Research Award for an Outstanding Poster/Oral Presentation by a Young Investigator, BLAST XIV, New Orleans, LA, 2017.
- Graduate College Conference Travel Award, UIUC, 2016.
- Graduated with honors (ranked 2nd/120), Department of Chemical Engineering, Sharif University of Technology, Iran, 2011.
- Singapore International Pre-Graduate Award, ICES, A*STAR, Singapore, 2010.

Professional Service

Reviewer for PLoS Computational Biology and PLoS One, 2018 – present.